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GEOG 485
Final Project

Remote Analysis of the Geology of Crater Lake

Introduction:

Throughout the course of this class, I have learned much about remote sensing and its various applications and tools. I have a clear understanding of: how satellites manipulate different bands of visible light wavelength in order to capture images, how different data collection methods like LIDAR, RADAR, and aerial photography work and how to manipulate their data differently, and how to operate different applications to process this data. For this reason, I wanted to structure my project around the discussion of data analysis in remote sensing. This includes knowing how it works, what kind of data to look for, and how to use tools to reach a desirable outcome image.

My initial draw to this class came from a place of personal interest and curiosity; I didn't have an exact goal for the outcome. However, by possessing the knowledge about remote sensing and having the skills to analyze satellite and other types of remote sensing data, I am now able to apply these tools in future professional applications. This project has allowed me to expand my skillset, as well as further my current understanding. Because of this interest, I wanted to chose a location that I had a high draw to – Crater Lake, Oregon.

When considering a research topic, I thought back to lab 1 where we were tasked to explore a place of our choice using different aerial image sources. For that assignment, I originally wanted to explore how to manipulate satellite imagery to show the types/classifications of the country rock in a region, however I soon realized that I didn't yet have the skillset needed to perform this type of analysis, so I stuck to an easier approach. Next, I also needed to pick a place. I love the geology of Crater Lake, so I chose to focus there. I knew I needed to find a satellite with the right band types to capture this, then choose what I wanted to show, before I could start processes sing the image to reflect that. This started the research portion of this project.

Methods:

My first step was to figure out what bands I would need to focus on in order to zero out rock types on the ground. From various sources I found ASTER is the better option over Landsat to look for imagery data, because it includes 14 spectral bands—and its six SWIR bands and five TIR bands—which "capture diagnostic mineral absorption and emissivity features that Landsat-8's broader, fewer SWIR and TIR bands cannot resolve." (Geoscience Australia, 2014). This finer spectral detail enables more accurate mapping of lithologies and hydrothermal alteration. After doing some searching on EarthExplorer and NASA websites, I collected satellite images from Landsat-8 and ASTER. ASTER is better for use for my purpose (NASA) so I started there.

ASTER required me to use a processing tool in QGIS to adjust the projection of the image. Due to this necessary manipulation of each band, my resulting image came out very grainy and static-like. While I attempted to research more about how to fix this, I did not yet have the understanding needed to do this. Thus, I turned to Landsat-8.

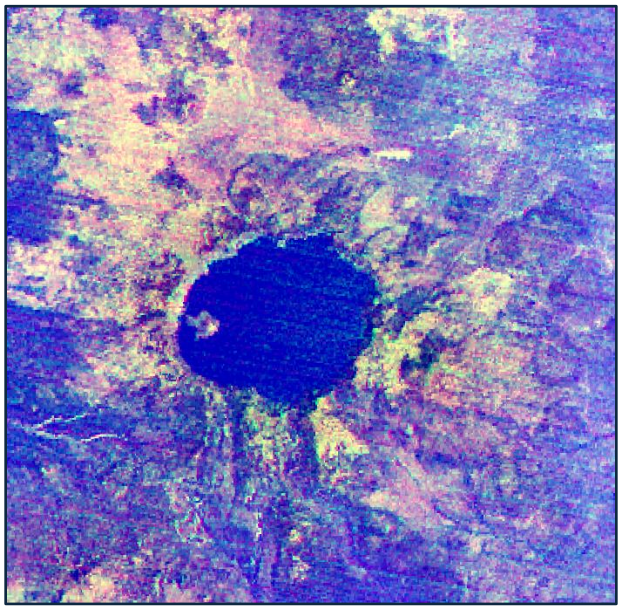

Figure 1: Crappy ASTER image result

I had two different images from the Landsat-8 image set, they are the same location just different times so I just chose one to use. I tried a bunch of different band combinations; with the basic idea following that I needed SWIR (Shortwave Infrared), which is critical for detecting minerals such as carbonates, clays, and quartz, and TIR (Thermal Infrared): Ideal for mapping

silicates and igneous rocks. I would use band combinations 7-6-4 and 7-5-3 for visually separating mafic vs felsic rocks (Hossain et. al, 2017).

Because of the volcanic nature of Crater Lakes formation and region, the surrounding rock is mostly felsic meaning it is rich in silicas. This is a trait common to igneous rocks. This is why I knew I needed to target these bands because of the radiance and reflexivity of felsic vs mafic rocks. I made up a couple different images in order to show the same things – with slight tweaks for each. This is the image I chose to highlight for this project:

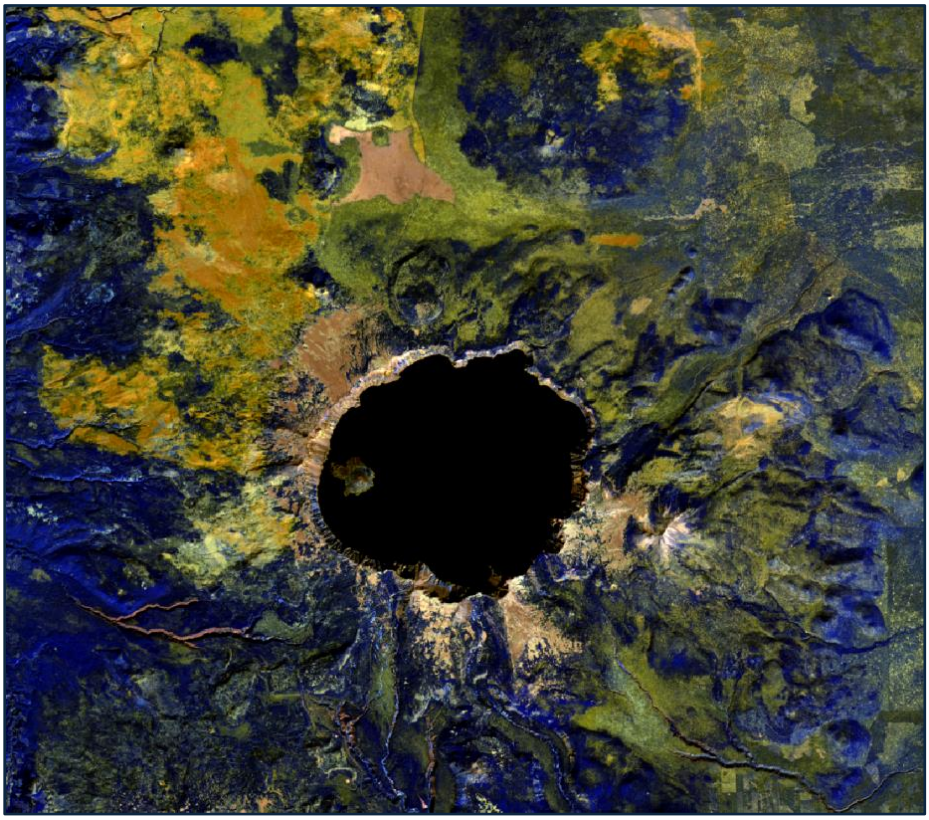

Figure 2: Rock type using configuration of Landsat-8 bands 7-5-4

I used the technique from this paper (step-by-step guide.pdf). It was stated that this combination of SWIR2, NIR, Red (B7, B5, B4) emphasizes iron-oxide and mafic/felsic contrasts; fresh mafic units often look darker/bluer and altered/felsic lighter/pinker. However, another combination I did consider was bands 3-4-7, where diorite and gabbro are displayed in bluish green, actinolite schist in green, basalt in black, biotite granodiorite gneiss in white, and tonalite and granodiorite in violet (JKSUS, 2020).

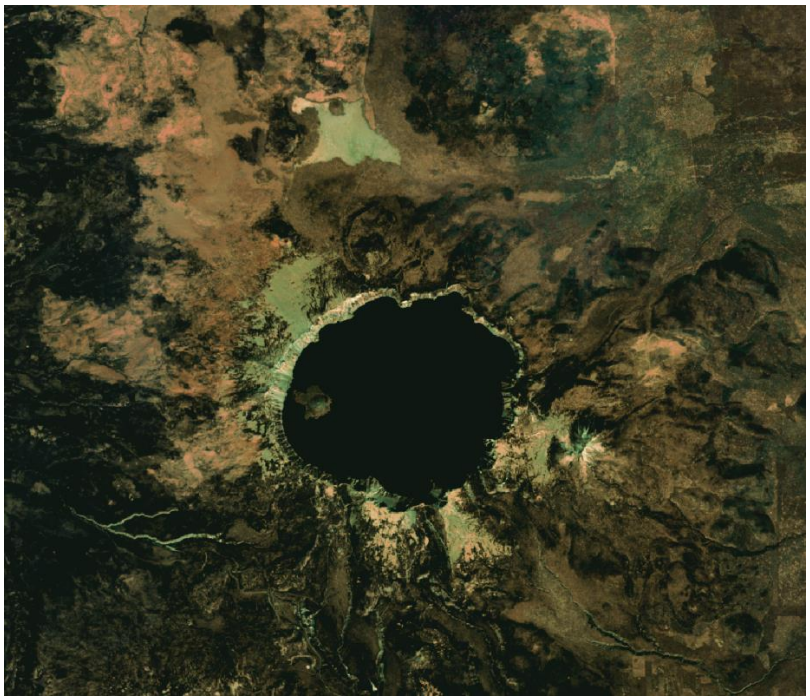

Figure 3: Landsat-8 bands 3-4-7

Results:

It was hard for me to pick one of these images to represent my data analysis while trying to answer my research question. The consensus from my analysis was that geological features of this area were highlighted using the 3–4–7 pan-sharpened color composite image of Landsat 8. Generally, these color combinations exhibit color variations corresponding to various geological units in the study area. This method provided the best results among all the methods I with respect to discriminating certain mafic vs felsic rock types, when comparing an image of know rock types of Crater Lake (Kamal et. al, 2023). This tactic is typically used by physically going to the research site and collecting evidence and maps of rock types. I wanted to be able to do this without needing physical evidence from the site. While this isn't as accurate and requires much more guess work, knowing how to analyze the data to get an intelligible output is the most important part.

This is another reason, along with the previously mentioned, being able to use the ASTER data accurately would have been the better route to take. All this being true, I ended up using Landsat-8 over ASTER for my final resolution. While this isn't the preferred method, the steps that needed to be taken in order to process ASTER data to be usable, is outside the scope

of my current skillset. This limitation left my result being less accurate, but overall, still a productive use of the skills I do have.

Discussion:

This project has provided an excellent opportunity to delve deeper into the tools and techniques involved in remote sensing data analysis, particularly in the context of geological mapping. The aim of this project was to explore different ways of creating an image showing the different mineral composites of the Crater Lake region. Through the course of the project, I faced numerous challenges, particularly with processing and interpreting the data, but also encountered significant learning experiences that will guide future work in remote sensing.

One of the primary lessons I learned through this project is the importance of selecting the appropriate dataset for the desired outcome. In theory, the ASTER satellite imagery would have been ideal for mapping geological features, due to its fine spectral resolution. As noted in my methods section, ASTER is particularly well-suited for this type of analysis because it can more accurately capture the diagnostic mineral absorption features that are critical for distinguishing lithologies.

Looking forward, I recognize that gaining proficiency in data manipulation and interpretation is crucial for the successful application of remote sensing in geological studies. The ability to manipulate multispectral images to detect subtle geological features is a powerful tool that can be applied to a wide range of fields, including environmental monitoring, mineral exploration, and land-use planning. As I continue to refine my skills, I aim to tackle more advanced datasets, and eventually return to ASTER imagery with a deeper understanding of image processing techniques to achieve more accurate and detailed results.

In conclusion, this project has not only enhanced my understanding of remote sensing tools and techniques but also provided me with a clearer vision of how to apply these tools in future professional contexts. Despite the challenges I faced, I gained invaluable experience in working with satellite imagery, and I am confident that this foundational knowledge will allow me to tackle more complex analyses in the future. The key takeaway from this research is the importance of both selecting the right data and developing the necessary skills to process and

interpret that data effectively. With continued practice and a deeper understanding of remote sensing techniques, I look forward to exploring the many applications of this field in the future.

Resources:

1. Esri. (n.d.). *Band combinations for Landsat 8*. ArcGIS Blog. Retrieved December 10, 2025, from https://www.esri.com/arcgis-blog/products/product/imagery/band-combinations-for-landsat-8
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3. Kamal, M. M., & Islam, M. N. (2023). Rock type discrimination using Landsat-8 OLI satellite data in mafic-ultramafic terrain. *ResearchGate*. Retrieved December 10, 2025, from https://www.researchgate.net/publication/374448226_Rock_type_discrimination_using_Landsat-8_OLI_satellite_data_in_mafic-ultramafic_terrain
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6. Hossain, M. S., & Kamal, M. M. (2017). Landsat-8 band ratio 7-5-5-4-6-7 in RGB of the study area. *ResearchGate*. Retrieved December 10, 2025, from https://www.researchgate.net/figure/Landsat-8-band-ratio-7-5-5-4-6-7-in-RGB-1-of-the-study-area_fig3_342096590
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9. Geoscience Australia. (2014). *ASTER data: A tool for identifying minerals and hydrothermal alteration*. Retrieved December 10, 2025, from https://www.ga.gov.au/bigobj/GA7833.pdf